

DS 340 Final Project Proposal

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Elevator Pitch : Develop a deep learning model that identifies the weather conditions from a given image and recommends appropriate clothing based on the detected weather.

Methods :

- **Convolutional Neural Network (CNN) :**
 - Use a Convolutional Neural Network (CNN) to analyze images of different weather conditions (e.g., sunny, rainy, snowy) and classify the type of weather. After identifying the weather, implement a recommendation system that suggests clothing based on predefined rules for each weather category. For instance, if the model detects "rainy," it might recommend a raincoat and umbrella.
- **Data Augmentation for Improved Generalization :**
 - Weather conditions in real-world images can vary due to different angles, lighting, and obstructions. Data augmentation allows us to artificially increase the diversity of the training data by applying transformations, thus helping the model generalize better. We will apply augmentation techniques such as random rotations, horizontal flips, zooming, and brightness adjustments to simulate real-world variability in weather conditions. This will help prevent overfitting and improve model robustness.
- **Ensemble Learning for Enhanced Accuracy :**
 - Using multiple models instead of a single CNN can reduce overfitting and increase accuracy by averaging out the predictions. We will create an ensemble of different CNN models (e.g., VGG16, ResNet, and custom CNN) or an ensemble of the same model trained with different subsets of the data. The predictions from the individual models will be averaged to make a final decision, ensuring that the system is more robust and less prone to errors.

Data Source :

- **Weather Image Train Datasets:**
 - **Snowy :** <https://universe.roboflow.com/weatherdetection/snowy-weather/dataset/1>
 - **Rainy :** <https://universe.roboflow.com/pron-w14aj/-lfbrw/dataset/14>
 - **Sunny :** <https://images.cv/dataset-categories/weather>

- **Foggy :**
<https://universe.roboflow.com/detection-objects/fog-detection-zfwki/dataset/2>
- **Cloudy :**
<https://www.kaggle.com/datasets/pratik2901/multiclass-weather-dataset>
- **Weather Image Test Dataset :**
<https://www.kaggle.com/datasets/ammaraifaifi/5class-weather-status-image-classification>

Code resources :

- **OpenAI Codex or ChatGPT for Research on Model Architectures**
 - Why : Codex (via ChatGPT) can assist in researching and exploring new CNN architectures, optimization strategies, and best practices in image classification tasks.
 - How : We will use these tools to explore alternate architectures, test new ideas, and solve complex coding challenges that require insight beyond standard documentation.
 - Primary Source : [OpenAI Codex](#), integrated within ChatGPT.
- **TensorFlow/Keras Documentation and Tutorials**
 - Why : TensorFlow and Keras are the main frameworks we will use for implementing the Convolutional Neural Network (CNN) model, transfer learning, and potentially other deep learning techniques.
 - How : We will refer to TensorFlow's official documentation and example code for setting up CNNs, performing image preprocessing, data augmentation, and implementing transfer learning.
 - Primary Source : TensorFlow Documentation and Keras Examples.

What's New :

- **Data Augmentation Techniques Specific to Weather**
 - **Contribution:** We will implement custom data augmentation strategies tailored to weather images. For example, we will design augmentations that simulate various real-world weather effects, such as cloud cover changes, variations in snow coverage, or fog density.
 - **Effort:** This involves coding data augmentation pipelines that simulate different weather-related visual effects. These augmentations will help improve the generalizability of the model by teaching it to recognize weather conditions under various circumstances.
 - **Why It's New:** While general data augmentation techniques are common, our weather-specific augmentations will be original and designed for this particular project.

Plan :

- **First Milestone (due by October 30) :** We will gather more weather image datasets and apply preprocessing techniques such as normalization and data augmentation. This step ensures that all images are ready for training.
- **Second Milestone (due by November 20) :** We will train multiple CNN models (ResNet, VGG16, and a custom CNN) and implement an ensemble learning approach to enhance classification accuracy.
- **Third Milestone (due by November 30) :** The weather classification system will be integrated with a rule-based clothing recommendation system. This includes preparing the final paper and ensuring model optimization before the December 9 deadline.

Proposed Demonstration or Evaluation :

- **Weather Classification Accuracy:**
 - We will evaluate the performance of the weather classification CNN on both training and testing datasets. Accuracy and F1-score will be used as the primary evaluation metrics to assess how well the model generalizes to unseen data.
 - We will showcase how well the ensemble model (average prediction of multiple CNNs) improves accuracy over individual models.
- **Clothing Recommendation System:**
 - The recommendation system will be demonstrated by feeding images of different weather conditions and showing the corresponding clothing recommendations (e.g., recommending a raincoat for rainy images).
 - We will measure the correctness of recommendations based on predefined clothing rules for each weather type and evaluate them qualitatively.

Experiments :

- **Data Augmentation Variations:**
 - **Experiment 1: Test the impact of different data augmentation strategies on model performance.**
 - Settings: Apply different levels of augmentation. Ex) rotations (0–90 degrees), brightness changes ($\pm 30\%$), and horizontal flips.
 - Measurement: We will track how these augmentations affect overfitting, validation accuracy, and model robustness, especially on varied test data. We expect augmented models to perform better in real-world scenarios.
- **Ensemble Learning Variations:**
 - **Experiment 2: Compare ensemble methods vs. individual CNN models.**

- Settings: We will compare the performance of individual models (VGG16, ResNet, custom CNN) against ensemble models, where predictions from these networks are combined (average or majority voting).
- Measurement: Metrics such as accuracy, precision, recall, and F1-score will be used to determine if the ensemble method significantly improves overall model performance.